

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Area and volume

01

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

Square A has side lengths that are 246 times the side lengths of square B. The area of square A is k times the area of square B. What is the value of k ?

- A. 60,516
- B. 492
- C. 246
- D. 123

Q2

GRID-IN

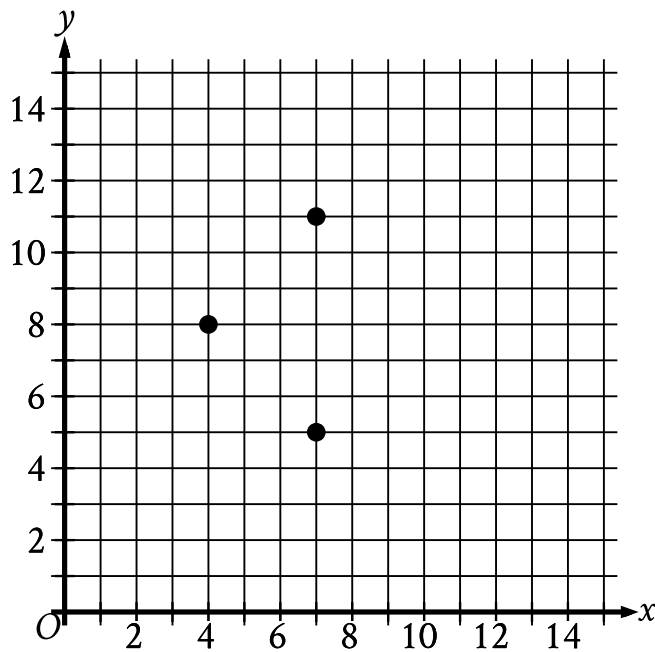
GEOMETRY AND TRIGONOMETRY

A triangular prism has a height of 8 centimeters cm and a volume of 216 cm^3 . What is the area, in cm^2 , of the base of the prism? (The volume of a triangular prism is equal to Bh , where B is the area of the base and h is the height of the prism.)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- From left to right, the 3 points have the following coordinates:
 - (4 comma 8)
 - (7 comma 11)
 - (7 comma 5)

The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

- A. 3
- B. 6
- C. 7
- D. 9

A manufacturing company produces two sizes of cylindrical containers that each have a height of 50 centimeters. The radius of container A is 16 centimeters, and the radius of container B is 25% longer than the radius of container A. What is the volume, in cubic centimeters, of container B?

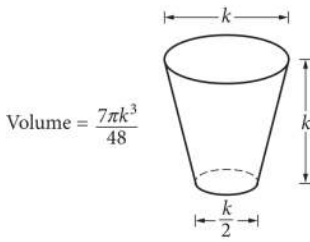
- A. $16,000\pi$
- B. $20,000\pi$
- C. $25,000\pi$
- D. $31,250\pi$

Square X has a side length of 12 centimeters. The perimeter of square Y is 2 times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

- A. 6
- B. 10
- C. 14
- D. 24

Q6

GEOMETRY AND TRIGONOMETRY



The glass pictured above can hold a maximum volume of 473 cubic centimeters, which is approximately 16 fluid ounces. What is the value of k , in centimeters?

- A. 2.52
- B. 7.67
- C. 7.79
- D. 10.11

Q7

GRID-IN

GEOMETRY AND TRIGONOMETRY

A circle has a circumference of 31π centimeters. What is the diameter, in centimeters, of the circle?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN

GEOMETRY AND TRIGONOMETRY

A right circular cylinder has a volume of 377 cubic centimeters. The area of the base of the cylinder is 13 square centimeters. What is the height, in centimeters, of the cylinder?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A triangle has a base length of 56 centimeters and a height of 112 centimeters. What is the area, in square centimeters, of the triangle?

- A. 168
- B. 1,568
- C. 3,136
- D. 6,272

Q10

GRID-IN

GEOMETRY AND TRIGONOMETRY

The length of each edge of a box is 29 inches. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square inches, of this box without a lid?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.